

Lecture 09 - Texturing

This lecture treats textures as sampled data for shading, not merely images glued onto objects. It connects UV coordinates, sampling, filtering, mipmaps, OpenGL texture state, and advanced texture uses.

Source chunk	course_text_parts/03_lectures/09_texturing.txt
Extracted slide pages	133
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Slide walkthrough

Each row gives a short reading comment for one extracted slide page. Use it to turn sparse slide bullets into a human-readable path through the lecture.

Page	Slide title	How to read it
1	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
2	Textures: Sampled Data for Shading	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
3	Texturing Example	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
4	Texture Mapping	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
5	Why Textures Matter	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
6	Textures Are More Than Color Images	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
7	Texturing Demo	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
8	Overview	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.

Page	Slide title	How to read it
9	9.1 Texture Objects, Texels, Formats, and Color Space	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
10	From Image File to GPU Texture Object	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
11	Texels, Channels, and Meaning	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
12	Texture Targets and Dimensionality	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
13	Internal Texture Formats	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
14	Numeric Texture Representations	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
15	Color Textures vs. Data Textures	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
16	2D Texture Access	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
17	Bilinear Interpolation	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
18	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
19	Texture Memory Footprint	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
20	Texture Compression and Asset Pipelines	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
21	Steps Involved in Texturing	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
22	9.2 Texture Coordinates, UV Mapping, and Wrapping	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
23	Texture Coordinate System 1/2	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.

Page	Slide title	How to read it
24	Texture Coordinate System 2/2	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
25	Texture Coordinate Specification	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
26	Disc Parametrization	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
27	Cylinder Parametrization	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
28	Two-Part Mapping 1/2	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
29	Two-Part Mapping 2/2	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
30	Two-Part Mapping Example	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
31	Projection Calculation Variations	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
32	Texture Coordinate Transformation	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
33	Demo - Texture Coordinate Transformation	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
34	Height Field Texturing Example	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
35	9.3 Sampling, Filtering, Mipmaps, and Anisotropy	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
36	Forward vs. Inverse Mapping	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
37	Forward Mapping	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
38	Inverse Mapping 1/2	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.

Page	Slide title	How to read it
39	Inverse Mapping 2/2	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
40	Magnification vs. Minification	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
41	Nearest vs. Linear Interpolation	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
42	Minification Filtering Requirements	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
43	Demo - Magnification Filtering: Nearest vs Linear	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
44	MIP Mapping (multum in parvo = 'much in little')	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
45	MIP Map Pyramid 1/2	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
46	MIP Map Pyramid 2/2	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
47	MIP Mapping Memory	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
48	MIP Mapping Procedure	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
49	LoD Calculation 1/2	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
50	LoD Calculation 2/2	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
51	Demo - Minification, Mipmaps, and Mipmap Level Visualization	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
52	Anisotropic Filtering	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
53	Demo - Anisotropic Filtering	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.

Page	Slide title	How to read it
54	9.4 Modern OpenGL Texture Pipeline	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
55	OpenGL Texturing	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
56	OpenGL Minification & Magnification 1/3	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
57	OpenGL Minification & Magnification 2/3	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
58	OpenGL Minification & Magnification 3/3	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
59	Demo - Texture Units and Sampler Uniforms	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
60	Texture Wrapping 1/4	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
61	Texture Wrapping 2/4	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
62	Texture Wrapping 3/4	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
63	Texture Wrapping 4/4	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
64	Texel Value Usage	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
65	Texturing in GLSL Shaders 1/3	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
66	Texturing in GLSL Shaders 2/3	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
67	Texturing in GLSL Shaders 3/3	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
68	9.5 Normal Mapping	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.

Page	Slide title	How to read it
69	Motivation	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
70	Mesh Compression	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
71	Normal Mapping 1/2	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
72	Normal Mapping 2/2	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
73	Normal Maps	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
74	Tangent Space	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
75	TBN Matrix	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
76	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
77	Tangent Vector Derivation 1/2	Read this as continuous-to-discrete conversion. Ask which samples are covered and which attributes are interpolated.
78	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
79	Tangent Vector Derivation 2/2	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
80	Untitled slide	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
81	Observations	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
82	Variations	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
83	9.6 Environment Mapping	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.

Page	Slide title	How to read it
84	Environment Mapping Motivation	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
85	Environment Mapping Process	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
86	Environment Mapping Assumptions	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
87	Environment Mapping 2/2	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
88	Demo - Cube-map Reflection Vector	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
89	Environment Mapping Types 1/2	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
90	Environment Mapping Types 2/2	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
91	Spherical Mapping 1/2	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
92	Spherical Mapping 2/2	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
93	Spherical Environment Mapping	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
94	Sphere Map Distortion	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
95	Demo - Environment Map Distortion	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
96	Sphere Map Interpolation	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
97	Sphere Map Coordinates	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
98	Cube Environment Mapping	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.

Page	Slide title	How to read it
99	Cube Mapping Example 1/2	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
100	Cube Mapping Example 2/2	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
101	Cube Mapping Distortion	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
102	Cube Map Texture Coordinates	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
103	Environment Mapping in OpenGL	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
104	Cube Map Texture Object	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
105	Sampling Direction	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
106	Environment Mapping in GLSL 1/2	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
107	Environment Mapping in GLSL 2/2	Read this as camera geometry. Track how view-space positions become clip, normalized, and screen coordinates.
108	Practical Cube-Map Issues	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
109	<code>1 glEnable(GL_TEXTURE_CUBE_MAP_SEAMLESS);</code>	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
110	Cube Mapping Pros and Cons	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
111	Spherical Environment Mapping Assumptions	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
112	Spherical Environment Mapping Conclusions	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
113	9.7 3D Textures and Volume Rendering	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.

Page	Slide title	How to read it
114	3D Texturing: Relevant Topics	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
115	3D Texture as a Voxel Grid	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
116	Object Space to Texture Space	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
117	3D Texture Access	Read this as a coordinate-space step. Name the input space, the matrix or operation, and the output space.
118	Trilinear Interpolation	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
119	$f = (1 - \alpha) \cdot f + \alpha \cdot f$	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
120	3D Texture Memory Footprint	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
121	3D Mipmaps and Filtering	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
122	Solid Texturing	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
123	Volumetric Data as Texture Data	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
124	Transfer Functions	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.
125	Volume Rendering with 3D Textures	Read this as sampled data. Identify the texture coordinates, sampling rule, filtering mode, and shader interpretation.
126	Front-to-Back Alpha Compositing	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
127	Ray Marching Volume Rendering	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
128	Gradients and Volume Lighting	Read this as local shading. Name the normal, light vector, view vector, material term, and where the computation happens.

Page	Slide title	How to read it
129	Quality and Performance Trade-Offs	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
130	3D Texturing Takeaways	Read this as an occlusion decision. Decide whether the method works in object space, image space, or per-fragment depth space.
131	■ Textures are sampled GPU data sources used by shaders	Read this as pipeline state and data flow. Ask which OpenGL object or shader stage owns the data at this point.
132	Literature and other sources used in this chapter	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?
133	■ Text Books	Read this slide by connecting the bullet terms causally: what problem is being solved, what data is used, and what output is produced?

09.1 Texture Objects, Texels, Formats, and Color Space

Source location: Section 9.1

Commentary

A texture object stores sampled data on the GPU. A texel is one stored texture sample. The texture format decides what channels and numeric representation the samples use. Color space matters because color values may be stored nonlinearly, especially for sRGB images.

The key shift is to stop thinking of texture as only a picture. Textures can store color, normals, depth, lookup data, environment information, or volume data. The shader decides how the sampled values are interpreted.

Mental model

A texture is a GPU-accessible sampled data field.

Check yourself

Why can the same texture mechanism store both color maps and normal maps?

09.2 Texture Coordinates, UV Mapping, and Wrapping

Source location: Section 9.2

Commentary

Texture coordinates map surface points to locations in texture space. UV coordinates are usually interpolated across primitives during rasterization and then used by the fragment shader to sample a texture.

Wrapping rules decide what happens outside the normal coordinate range. Repeat, clamp, mirrored repeat, and border behavior are not visual afterthoughts; they define the sampling function outside the base domain.

Mental model

UVs are the address system that lets a fragment look up texture data.

Check yourself

What artifact might appear if a wrapping mode is wrong at the edge of a surface?

09.3 Sampling, Filtering, Mipmaps, and Anisotropy

Source location: Section 9.3

Commentary

Sampling turns continuous texture coordinates into values from discrete texels. Nearest filtering selects a nearby texel and can look blocky. Linear filtering blends nearby texels and looks smoother. Minification is harder because many texels may map to one pixel.

Mipmaps store prefiltered lower-resolution versions of a texture. They reduce aliasing and shimmer when textures are seen at small scale. Anisotropic filtering improves quality when a texture is viewed at a steep angle where footprint shape is elongated.

Mental model

Filtering reconstructs values; mipmaps choose an appropriate scale before reconstruction.

Check yourself

Why does a distant checkerboard shimmer without mipmapping?

09.4 Modern OpenGL Texture Pipeline

Source location: Section 9.4

Commentary

In modern OpenGL, textures are represented by texture objects, bound to texture units, connected to sampler uniforms, and accessed in shaders. The shader does not sample a filename; it samples a bound texture through a sampler.

Texture bugs often come from state mismatches: wrong active texture unit, wrong sampler uniform, missing mipmaps for a mipmap filter, wrong wrap mode, or wrong internal format.

Mental model

OpenGL texturing is a chain: object data -> texture unit -> sampler uniform -> shader lookup.

Check yourself

Why can a texture appear black when the shader code is mathematically correct?

09.5 Normal Mapping

Source location: Section 9.5

Commentary

Normal mapping stores normal directions in a texture so that lighting can vary at a finer scale than the geometry. The surface may have few triangles, but the shader receives detailed normals per fragment.

The hard part is coordinate space. Normal maps are often defined in tangent space, so the shader must transform or interpret them with the correct tangent, bitangent, and normal basis.

Mental model

Normal mapping changes the lighting normal, not the actual mesh silhouette.

Check yourself

Why does normal mapping not change the geometric outline of an object?

09.6 Environment Mapping and 3D Textures

Source location: Sections 9.6-9.7

Commentary

Environment mapping uses textures to represent surrounding illumination or reflections. A direction, not a surface UV alone, can be used to sample an environment map. This is a texture lookup driven by view/reflection geometry.

3D textures and volume rendering extend the same sampled-data idea into three dimensions. Instead of sampling a 2D image, the shader samples a volume. This is useful for data such as medical scans, density fields, or procedural volumetric effects.

Mental model

Textures are general sampled data; the coordinate dimensionality depends on the problem.

Check yourself

What coordinate type would you use to sample a 3D texture?

End-of-lecture summary

If you remember only one thing from Lecture 09, remember this: This lecture treats textures as sampled data for shading, not merely images glued onto objects. It connects UV coordinates, sampling, filtering, mipmaps, OpenGL texture state, and advanced texture uses.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.